INCH-POUND

MIL-PRF-1/4J
14 June 2007
SUPERSEDING
MIL-E-1/4H
4 February 1981

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 5654W 1/

Inactive for new design
after 7 March 1997.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and MIL-PRF-1.

DESCRIPTION: Pentode, miniature, RF sharp cutoff.

Outline - - - 5-1 (EIA).
Base - - - E7-1.
Envelope - - - T5-1/2.
Cathode - - - Coated unipotential.

Base connections:

Pin No.	- - -	1	2	3	4	5	6	7
Element	- - -	g1	k, g3 int sd	h	h	a	g2	k, g3 int sd

ABSOLUTE RATINGS:

Parameter:	Ef	Eb	Ec1	Ec2	Ehk	Rg1	Ic1	Ik	Pp	Pg2	TE	Alt
Unit:	V	V dc	V dc	V dc	v	MegΩ	mA dc	mA dc	W	W	°C	ft
Maximum:	6.9	200	0, -55	155	100	0.1	1.0	20	1.65	0.55	165	(See note 1)
Minimum:	5.7	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -
Test conditions:	6.3	120	-2	120	0	- - -	- - -	- - -	- - -	- - -	- - -	- - -

GENERAL:

Qualification: Not required.
Reliable tube.

1/ Formerly tube type 5654

AMSC N/A FSC 5960

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TABLE I. Requirement or test.

MIL-STD-1311 method	Requirement or test	Conditions	Acceptance level (see note 11)	Symbol	Limits		Unit
					Min	Max	
	Conformance inspection, part 1						
1301	Heater current		0.4	If	160	190	mA
1336	Heater-cathode leakage		0.4	Ihk	- - -	10	µA dc
1266	Total grid current	Rg = 0.5 MegΩ (see notes 2 and 3)	0.4	Ic1	0	-0.1	µA dc
1256	Electrode current (1) (anode)	See note 2	0.4	Ib	5.0	11.0	mA dc
1256	Electrode current (2) (anode)	Ec1 = -10 V dc; Rp = 0.1 MegΩ	0.4	Ib	- - -	200	µA dc
1256	Electrode current (screen)		0.4	Ic2	0.8	4.0	mA dc
1306	Transconductance (1)		0.4	Sm	3,800	6,200	µmhos
1201	Short and discontinuity detection		0.4	- - -	- - -	- - -	- - -
	Conformance inspection, part 2						
1211	Insulation of electrodes		2.5	- - -	- - -	- - -	- - -
1256	Electrode current (3) (anode)	Ec1 = -5.5 V dc	2.5	Ib	5.0	- - -	µA dc
1306	Transconductance (2)	Ef = 5.7 V	2.5	ΔSm Ef	- - -	15	%
1266	Grid emission	Ef = 7.5 V; Ec1 = - 45 V; Rg1 = 0.1 MegΩ (See note 4)	2.5	Isg	0	- 0.5	µA dc

See notes at end of table.

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TABLE I. Requirement or test - Continued.

MIL-STD-1311 method	Requirement or test	Conditions	Acceptance level (see note 11)	Symbol	Limits		Unit
					Min	Max	
	Conformance inspection, part 2 - Continued						
2201	Noise and microphonics	Ef = 6.3 V ac; Ehk = 0; Ecc2 = 200 V dc; Ec1 = 0; Ebb = 200 V dc; Ecal = 175 mV ac; Rk = 1,000 ohms; Rp = 0.1 MegΩ; Rg2 = 0.5 MegΩ; Cg2 = 2 μF	2.5	- - -	- - -	- - -	- - -
1331	Direct-interelectrode capacitance	Shield No. 316	6.5	Cgip Cin Cout	- - - 3.40 2.45	0.020 4.6 3.25	pF pF pF
1031	Low-frequency vibration	Rp = 10,000 ohms; 10 G; F = 40 Hz	6.5	Ep	- - -	200	mV ac
1041	Shock	450 G; Ehk = +100 V dc (see note 5)	- - -	- - -	- - -	- - -	- - -
1031	Vibration-fatigue	2.5 G; fixed frequency; F = 25 min, 60 max	6.5	- - -	- - -	- - -	- - -
- - -	Post-shock and vibration-fatigue test end points:						
1031	Low-frequency vibration		- - -	Ep	- - -	300	mV ac
1336	Heater-cathode leakage		- - -	Ihk	- - -	30	μA dc
1306	Transconductance (1)		- - -	Sm(1)	3,500	- - -	μmhos
1266	Total grid current		- - -	Ic1	0	- 0.2	μA dc

See notes at end of table.

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TABLE I. Requirement or test - Continued.

MIL-STD-1311 method	Requirement or test	Conditions	Acceptance level (see note 11)	Symbol	Limits		Unit
					Min	Max	
	Conformance inspection, part 2 - Continued						
1121	Base strain	See note 7	- - -	- - -	- - -	- - -	- - -
2126	Glass strain		2.5	- - -	- - -	- - -	- - -
1105	Permanence of marking		- - -	- - -	- - -	- - -	- - -
MIL-STD-1311 method	Requirement or test	Conditions		Symbol	Limits		Unit
					Min	Max	
	Conformance Inspection, part 3						
1506	Heater-cycling life	Ef = 7.5 V; Ehk = +135 V dc; EC1 = Ec2 = Eb = 0		- - -	- - -	- - -	- - -
- - -	Heater-cycling life test end point						
1336	Heater-cathode leakage			Ihk	- - -	20	μA dc
1516	Stability life	Eb = 150 V dc; Ec1 = 0; Ec2 = 125 V dc; Ehk; = +135 V dc; Rk = 130 ohms; Rg1 = 0.1 MegΩ; TA = room		- - -	- - -	- - -	- - -
- - -	Stability life-test end point						
1306	Change in trans-conductance (1) of individual tubes			ΔSmt	- - -	10	%

See notes at end of table.

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TABLE I. Requirement or test - Continued.

MIL-STD-1311 method	Requirement or test	Conditions	Symbol	Limits		Unit
				Min	Max	
	Conformance Inspection, part 3 - Continued					
1501	Intermittent life	Stability life-test conditions; TE = +165°C (min) (see note 8)	- - -	- - -	- - -	- - -
- - -	Intermittent life-test end points (1,000 hours):					
- - -	Inoperatives		- - -	- - -	- - -	- - -
1266	Total grid current		Ic1	0	-0.1	μA dc
1301	Heater current		If	160	196	mA
1306	Change in transconductance (1) of individual tubes		ΔSmt	- - -	30	%
1306	Transconductance (2)		ΔSmEf	- - -	20	%
1336	Heater-cathode leakage		Ihk	- - -	10	μA dc
1211	Insulation of electrodes		R	50	- - -	MegΩ

See notes at top of next page.

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TABLE I. Requirement or test - Continued.

NOTES:

1. See "reduced pressure (altitude) rating" and altitude, maximum peak voltage in MIL-PRF-1.

2. This test shall be performed at the conclusion of the holding period.

3. The insertion of a 1.0-megohm resistor in the grid circuit shall change the anode current by not more than 0.5 mA dc. This is an alternative test.

4. Prior to this test, tubes shall be preheated a minimum of 5 minutes operating at the conditions specified below. The 3-minute test shall not be permitted. Test at specified conditions within 3 seconds after preheating. Grid emission shall be the last test performed on the sample selected for the grid-emission test.

Ef	Ec1	Ec2	Eb	Rk	Rg1
V	V dc	V dc	V dc	Ohms	MegΩ
7.5	0	125	150	130	0.1

5. A grid resistor of 0.1 megohm shall be added; however, this resistor shall not be used when a thyratron-type short indicator is employed.

6. This test shall be conducted on the initial lot and thereafter on a lot approximately every 12 months. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. When one lot has passed, the 12-month rule shall apply.

7. Acceptance sampling procedure shall be in accordance with "Base-strain test, miniature sampling" (method 1121), except that data covered in "Acceptance and rejection criteria" shall be modified as follows:

   (a) Accepted if no defectives for class "A", "B", or "C" defects, respectively (see method 1121), or if no defectives are found in the sample.

   (b) Rejected if there are any defectives for class "A", "B", or "C" defects, respectively.

8. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (MIL-STD-1311, method 1226), will be satisfied if a tube having bogey Ib (±5 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.

9. The life-test sample shall consist of the lesser of 20 tubes or 10 percent of lot size and no tube failures shall be permitted.

10. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

11. This specification sheet utilizes an accept on zero defect sampling plan in accordance with MIL-PRF-1, table III.

Referenced documents. In addition to MIL-PRF-1, this document references MIL-STD-1311.

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Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Army - AR, CR4, MI
Navy - AS, CG, MC, OS
Air Force - 99

Preparing activity:
DLA - CC

(Project 5960-2007-023)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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